7. Johnson, Ann. *Best Practices Handbook on Asphalt Pavement Maintenance*. University of Minnesota Center for Transportation Studies Report MN/RC 2000–04, February 2000.
8. MS-2, *Mix Design Methods*. The Asphalt Institute, Lexington, KY.
9. Roberts, Freddy L., Kandhal, Prithvi S., Brown, E. Ray, Lee, Dah-Yin, and Kennedy, Thomas. *Hot Mix Asphalt Materials, Mixture Design and Construction*, NAPA Education Foundation, Lanham, MD.

CHAPTER 13

REHABILITATION METHODS FOR ASPHALT PAVEMENTS

Section 13.1 HMA Overlay over Asphalt Pavements
13.1.1 Considerations in Overlay Selection

Section 13.2 Preparing Pavements for Overlays
13.2.1 Local Weak Areas (Patching)
13.2.2 Leveling Courses
13.2.3 Crack Filling/Sealing
13.2.4 Widening
13.2.5 Drainage
13.2.6 Interlayer Techniques

Section 13.3 Mill and Inlay
13.3.1 Specifications
13.3.2 Inspection

Section 13.4 Hot In-Place Recycling
13.4.1 Pavement Assessment
13.4.2 Material Evaluation
13.4.3 HIPR Construction Processes
13.4.4 Inspection

Section 13.5 Cold-Mix Recycling
13.5.1 Introduction
13.5.2 Central Plant (CP) Cold Recycling
13.5.3 Stationary Facilities
13.5.4 Placement of the Mix

Section 13.6 Cold In-Place Recycling
13.6.1 General
13.6.2 Mix Design
13.6.3 Equipment and Methods
13.6.4 Placement of CIPR Materials
13.6.5 Compaction of CIPR Layers
13.6.6 Curing and Aeration
13.6.7 Wearing Course
13.6.8 Inspection

Section 13.7 Full-Depth Reclamation

13.7.1 Stabilization with Chemical Additives
13.7.2 Stabilization with Asphalt Materials
13.7.3 Construction of an FDR Pavement
13.7.4 Shaping and Compaction Procedures
13.7.5 Inspection

13.1 HMA Overlays over Asphalt Pavements

The most popular method of pavement rehabilitation for both hot mix asphalt (HMA) pavements and Portland cement concrete (PCC) pavements is the asphalt overlay. This chapter discusses the rehabilitation of HMA pavements, and Chapter 14, "Rehabilitation of Concrete Pavements with HMA" covers the use of HMA overlays for PCC pavements. Other options for resurfacing, such as surface treatments, are discussed in Chapter 12, "Asphalt in Pavement Maintenance". Properly designed and constructed, an asphalt overlay is a cost-effective method of correcting surface deficiencies, restoring user satisfaction, and increasing the load-carrying capacity of the roadway.

The first step in developing an asphalt pavement rehabilitation project is to conduct a detailed project-level pavement evaluation using the procedures detailed in Chapter 11, "Pavement Management." This information is needed for identifying available rehabilitation alternatives and for designing the project. In addition, the designer needs data to assess the structural adequacy of the pavement. This data is collected by sampling and testing the materials in the pavement, by conducting in situ testing, or by conducting deflection testing. In some situations, the project's physical constraints may require pavement reconstruction rather than overlay.

The three purposes for overlaying an asphalt pavement are the following:

- **Heavy structural overlay.** A heavy structural overlay is more than 5 inches (125 millimeters) thick. These overlays considerably improve the load-bearing capacity of the pavement and usually have the same design period as a newly constructed pavement.
- **Structural overlay.** A structural overlay is 3–5 inch (75–125 millimeters) thick. These overlays add strength to an old pavement and last longer than functional or maintenance overlays.
- **Functional or maintenance overlay.** A functional overlay is 0.75–3 inches (20–75 millimeters) thick. Functional overlays are intended to address ride quality, surface friction or other minor problems for pavements that are structurally sound. Thin HMA overlays are also a popular choice for preventative maintenance. A functional overlay may be placed after the surface has been cold milled to replace the material that was removed. This technique is commonly called "mill and fill," or referred to as an inlay.

13.1.1 Considerations in Overlay Selection

The pavement designer has a number of options available for rehabilitating an existing HMA pavement. There is a point at which the extent of

pavement deterioration makes it more cost effective to reconstruct or to full-depth recycle the pavement rather than to place a thick overlay. Practical constraints, such as vertical clearance limitations, numerous driveways or intersections, etc., could also make reconstruction the best alternative. For example, poor drainage coupled with a poor base and/or subbase support cannot be corrected by placing an overlay. Also, reconstruction or full-depth recycling may be the desired alternative when serious durability problems, such as stripping, are present in the asphalt pavement being overlaid. The problem layer must be corrected where stripping, or moisture damage, is apparent. In a majority of cases, the most realistic approach to this is removal and replacement.

In selecting the proper rehabilitation alternative, a number of items need to be considered. The goal is to choose the most cost-effective solution for a particular problem. A discussion of these items follows. Table 13.1 gives some general guidelines related to the causes and solutions for pavement rehabilitation.

Pavement Condition

Pavement condition is the most important consideration when deciding which rehabilitation process to use. A discussion of the different pavement distress types and mechanisms is provided in Chapter 11, "Pavement Management," and MS-17, *Asphalt Overlays for Highway and Street Rehabilitation*. A pavement in good condition—little surface distress and no evidence of any structural failure problems—is a good candidate for a functional rehabilitation, which might include a surface treatment, mill and fill overlay or hot-in-place recycling. A pavement in poor to failed condition requires an extensive rehabilitation, which might include cold-in-place recycling or full-depth reclamation prior to an overlay.

Construction Feasibility

An HMA overlay requires less traffic control than a reconstruction project because in most cases the overlay can be placed one lane at a time. But some pre-overlay techniques, such as full-depth reclamation, require that the roadway be closed or that its usage be significantly restricted. Therefore, some techniques in urban areas or on mountainous roadways with limited width may not be appropriate. In some situations, the lack of specific equipment in the project locality may limit the feasibility of a particular rehabilitation method. For example, it may not be economically feasible to mobilize specific equipment for a small project, such as a hot-in-place or cold-in-place recycling train. Overlay placement that increases the thickness of the existing pavement has the disadvantage of decreasing overhead clearances under bridges or utilities. Also, the overlay placement in an urban area may cause loss of curb reveal or problems at adjoining driveways and intersections. Any possible clearance problems must be considered in the initial design, and steps must be taken to solve them. Possible solutions are to reconstruct the section under the bridge or utility or to mill the existing roadway below the level of the curb line and cross streets. These steps must be taken before placing the overlay.

Table 13.1
Some Alternatives for Pavement Rehabilitation

Problem	Contributing Causes				Possible Rehabilitation Technique[1]					
	Structural Failure	Mix Composition	Temp or Moisture Changes	Construction	Surface Recycling	Thin Overlay	Open graded surface	Structural Overlay[4]	Structural Recycling[5]	Reconstruction[5]
Alligator cracking	X			X					X	X
Reflective cracking				X					X	
Shrinkage cracking		X	X		X				X	
Rutting	X			X		X[2]			X	X
Corrugation	X			X		X[3]			X	X
Depressions	X			X					X	X
Potholes	X	X	X	X					X	
Raveling		X		X	X	X	X	X		
Bleeding		X	X	X	X	X	X	X		
Loss of skid resistance		X		X	X	X	X	X		

[1] When distress = 40 percent of area [2] When distress is minor [3] Over milled surface [4] After proper surface preparation [5] If problem is extensive

Performance

A number of factors can lead to diminished asphalt overlay performance, including the following:

- Insufficient overlay thickness
- The use of the wrong type of HMA surface for the overlay
- Insufficient preparation of the existing pavement before the overlay
- Insufficient consideration of the potential for reflection cracking

Overlay Thickness

The overlay thickness must be designed for the anticipated traffic over the design period. It should be designed using the principles and procedures discussed in Chapter 6, "Thickness Design." The selection of the proper material for the overlay is discussed in Chapter 4, "Hot Mix Asphalt (HMA) Design."

Pre-overlay Preparation

Overlay performance is influenced by the amount, type, and severity of distress in the existing pavement. Extensive repair of the existing pavement will maximize the performance of the overlay. For example, replacing all areas that have medium-to-severe alligator cracking and improving all areas with poor drainage on a project will significantly increase the life of a pavement rehabilitation project. This is one approach to pre-overlay preparation. The other approach is to place an overlay of sufficient thickness to protect all the weak areas in the existing pavement structure. If an existing base or subbase layer has deteriorated, the designer increases the thickness of the overlay to account for the decreased strength of the underlying layers, instead of removing that layer. As the overlay thickness is increased, the applied loads are distributed over a larger area in the lower layers, thereby decreasing the stresses and deflections on the weak layer. The thickness required to adequately protect weak layers usually makes this approach economically unfeasible. Figure 13.1 shows the total repair cost and the total overlay cost plotted against the percentage of area repaired. When the percentage of the total area repaired increases, the cost of the overlay will decrease. But when these two costs are added together, the total cost of the project reaches a point at which it is minimized. When the total cost begins to increase relative to the percentage of the area repaired, the designer should consider reconstructing or using full-depth reclamation as more cost-effective design alternatives.

Reflection Cracking Control

Reflection cracking is the appearance of the pattern of joints or cracks in the old pavement at the surface of an overlay, and can be a serious problem to consider when choosing an approach for asphalt pavement rehabilitation. A complete discussion of the mechanism of reflection cracking in overlays of PCC pavement is provided in Chapter 14, "Rehabilitation

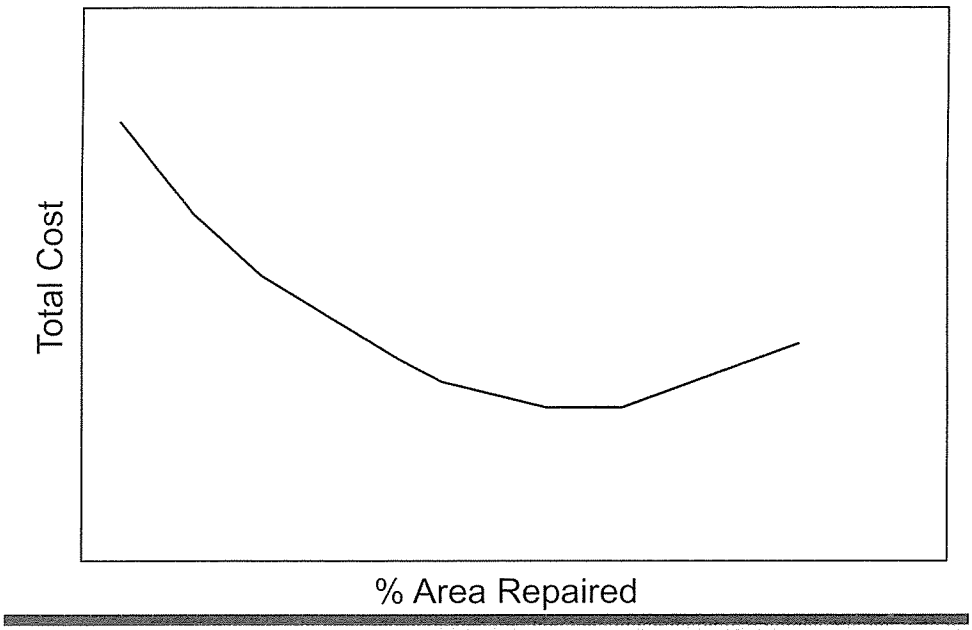


Figure 13.1 Overlay Cost and Repair Cost versus Percentage of Area Repaired

of Concrete Pavements with HMA." To control reflection cracking, the horizontal and vertical movements at existing cracks or joints must be reduced to the point where there is no significant tensile or shear strain immediately above those locations. Reflection cracking on asphalt pavements is usually reduced or controlled by using some type of interlayer or by using in-place recycling (cold, or full-depth reclamation).

13.2 Preparing Pavements for Overlays

13.2.1 Local Weak Areas (Patching)

Any structurally inadequate areas should be removed and replaced using proper patching techniques before placing the new overlay. Detailed visual inspection should be used to identify any load-associated distress such as alligator cracking. "Proofrolling," or observing the amount and the extent of deformation in the existing pavement while driving a loaded truck over the pavement surface, is a very quick and simple method of defining the structurally inadequate areas. These areas should be removed and patched following the procedures detailed in Chapter 12, "Asphalt in Pavement Maintenance."

13.2.2 Leveling Courses

If milling is not used to correct surface distresses, the placement of one or more leveling courses may be required to restore proper line and cross-section. A leveling course can be either a scratch course or a wedge course. Both are designed to fill in low spots on the existing pavement surface and to restore cross-slope.

Scratch courses can be placed with either a paving machine or a motor grader (Figure 13.2). They are thin courses of HMA used to fill in small undulations in the pavement surface. The HMA mix used for a scratch course generally has a small nominal maximum particle size (many times a sand mix or a 3⁄8 inch [4.75 millimeters] mix). Care should be taken that the aggregate in the mix used in a scratch coat not be so large that dragging of the mix with the motor grader or the paving machine results.

Leveling wedges are courses of HMA used to level depressions or correct cross-slope issues (for example, adding superelevation to an otherwise flat horizon curve) in an old pavement before additional layers of HMA are placed. Leveling wedges should be placed in lifts of not more than 3 inches (75 millmeters). Figure 13.3 illustrates several design considerations for constructing leveling wedges. The HMA can be placed with either a motor grader or a paving machine. Care should be taken in choosing the mix for a leveling course. The thickness of the leveling course should be at least three times the nominal maximum particle size for the mix being used.

13.2.3 Crack Filling/Sealing

A badly cracked pavement, where the type of cracking indicates a weak pavement structure or inadequate materials, should be repaired or

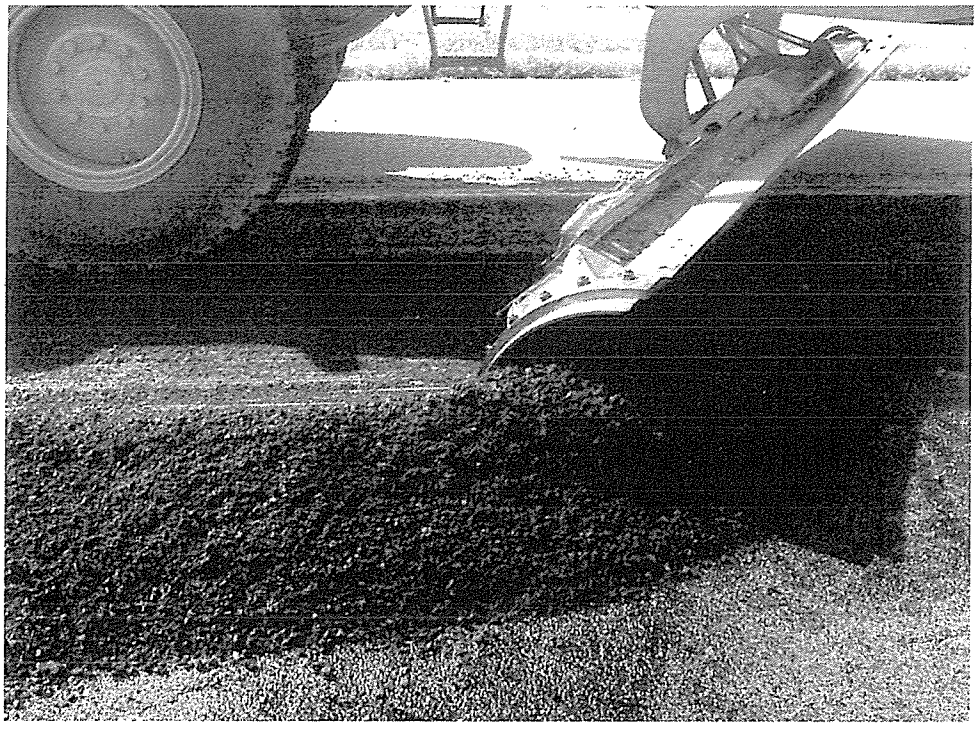


Figure 13.2 Placing a Leveling Course with a Motor Grader

replaced. Filling the cracks on these pavements will be of little to no value. The benefits of sealing or filling other cracks prior to an overlay will depend on the width of the cracks. If they are narrow, the crack-sealant materials will not enter the crack. If the cracks are less than ¼ inch (8 millimeters) wide, crack sealing is probably not appropriate. Wider cracks should be blown out with compressed air and cleaned of debris. The crack-sealing material should be placed in a clean and dry crack. Its level should be slightly lower than that of the surrounding surface and should not spill over the sides of the crack. Some agencies require that crack sealing operations be suspended at least six to twelve months prior to the placement of an overlay. The reason is that when the overlay is placed on a newly crack-sealed surface the pavement expands, closing the crack and pushing the filler material up and out of the crack, which will cause a bump in the overlaid surface. If the surface has extensive cracking, milling the surface before overlay can be very effective. The cracks are usually wider at the surface and milling off 1–2 inches of the surface can result in having much narrower cracks below the new overlay. These cracks are sealed with the tack coat during the construction process. A detailed discussion of crack sealing/filling is in Chapter 12, "Asphalt in Pavement Maintenance."

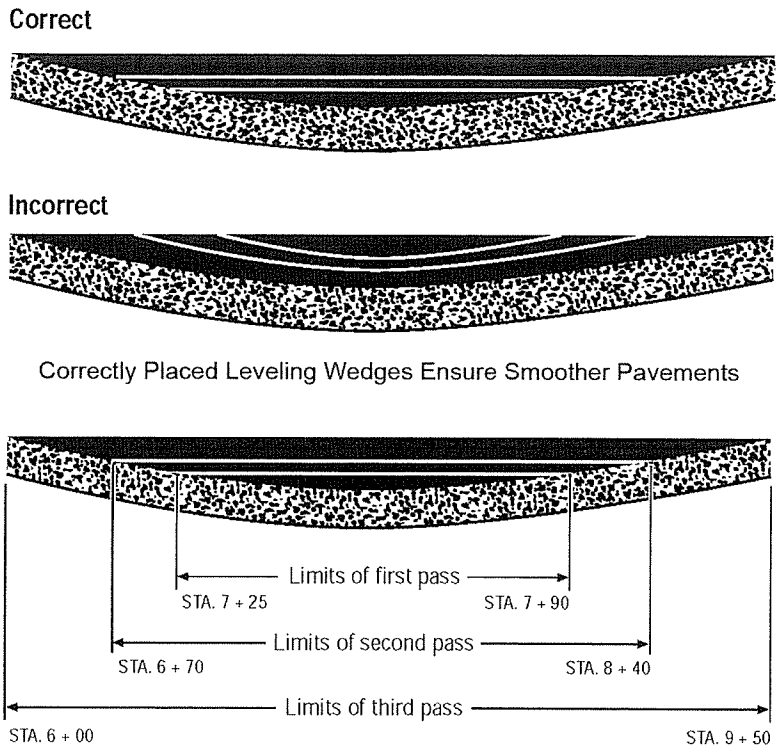


Figure 13.3 Leveling Wedge Construction

13.2.4 Widening

Pavement widening includes adding paved shoulders to an existing pavement or adding additional lanes. The thickness for the widened section should be designed and constructed to match the existing pavement plus the overlay pavement. The structural number (see Chapter 6 "Thickness Design") should be equal or exceed the structural number of the mainline pavement. The materials and the construction procedures used for widening are similar to those used for new HMA pavement construction. A geotechnical engineer should be consulted to evaluate the need for a subsurface longitudinal drainage system, and, if necessary, design one.

When widening a pavement, it is important to ensure that the additional pavement is tied to the existing pavement. This can be done by making a longitudinal saw cut along the existing pavement and then applying a

tack coat to the vertical face. This accomplishes two things: it tacks the new HMA to the existing roadway and helps to seal the longitudinal crack between the old and new pavement.

13.2.5 Drainage

If drainage problems are a contributing factor to the need for the overlay, they must be addressed. A frequent cause of failure is inadequate pavement and subgrade drainage. An assessment of the drainage-related pavement damage is critical in the development of the overlay design. Correcting a drainage problem can be as simple as cleaning or deepening the side ditches to permit rapid runoff of surface water, or the correction may be as complex as installing a longitudinal drainage system. Chapter 5, "Soils and Drainage for Asphalt Pavements," contains a discussion of the design and construction of a longitudinal drainage system.

13.2.6 Interlayer Techniques

Function of an Interlayer

An interlayer is a layer of material placed between an existing pavement (either HMA or PCC) and the overlay. The purpose of the interlayer is to minimize the reflection cracking in an overlay. Reflection cracking is the result of horizontal and vertical movements in the cracks in the underlying asphalt pavement. Reflection cracking results in increased roughness and infiltration of surface water into the underlying layers. This increase in moisture results in a weakening of the support for the pavement. The use of an interlayer will not always prevent reflection cracking, but it has been found that the proper use of an interlayer will:

- Retard the occurrence of the reflection cracking (delay the onset of the cracking)
- Reduce the severity of the crack when it does occur
- Provide a waterproofing membrane for the underlying layers
- Reinforce the asphalt overlay to reduce horizontal stresses in the asphalt layer

Geotextiles

Woven or non-woven synthetic geotextiles made of polypropylene, polyester, fiberglass, nylon, and combinations of these are used as interlayers for HMA pavements. There are many of these products on the market, and the engineer is encouraged to contract the local DOT and various suppliers for information on a specific product. They are placed directly on the existing pavement surface after a heavy tack coat is applied. The geotextile is rolled or brushed into the tack coat and an overlay is placed on top of the geotextile. Many agencies place the geotextile within the overlay. This requires two lifts of HMA and a minimum overlay thickness of 3 inches (75 millimeters).

The advantage of this approach is that the first lift or leveling course provides an unblemished surface and reduces the potential for the wrinkling of the geotextile during placement. Geotextiles should not be used where there is substantial horizontal or vertical movement at the joint or the crack in the existing pavement. The use of geotextiles in the pavement structure may preclude the use of any type of recycling technique in the future. Geotextiles placed on longitudinal joint cracking, where the horizontal movement of the existing pavement is small, have proven to be effective.

The specifications for geotextiles used when rehabilitating an asphalt pavement usually include the following details:

- **Weight** (oz/yd^2/gm/m^2). This is the quantity of geotextile needed to absorb a sufficient amount of tack coat to form a membrane.
- **Grab tensile strength** [pounds (Newtons)]. This property is important for preventing the geotextile from separating during the placement process and to ensure that the geotextile performs as a reinforcing layer.
- **Elongation at break** (percent). This is determined during the grab tensile test and is required to insure that the geotextile will not pull apart during placement.
- **Asphalt retention** [gallon/yard^2(litre/meter^2)]. This is an indication of how much asphalt is necessary to saturate the geotextile and make a bond. Different geotextiles will absorb different amounts of asphalt, depending on the weight and thickness of the geotextile.
- **Miscellaneous.** The specifications will require that the geotextile be kept in a plastic sleeve before placement to prevent ultraviolet-ray damage to the geotextile; and that the melting point of the geotextile is at least 300°F (150°C) to prevent the geotextile from melting during construction.

Table 13.2 gives typical values for the above items. Designers should check with the DOT or the geotextile manufacturer for their recommendations on the specified product being considered.

Any structural repairs (such sections with alligator cracking, base failure, or subgrade failure) in the existing pavement should be completed prior

Table 13.2
Typical Paving Nonwoven Geotextile Specifications

Specification	Requirement
Weight ASTM Designation: D 5261	4 oz/yd^2 (135 gm/m^2)
Grab tensile strength ASTM Designation: D 4632	1 in. grip @ 100 lb (25 mm grip @ 450 N)
Elongation at break, percent min. ASTM Designation: D 4632	50
Asphalt retention by geotextile, residual minimum ASTM Designation: D 6140	0.2 gal/yd^2 (0.9 l/m^2)

to placement of a geotextile interlayer. A smooth surface is necessary for ensuring that the geotextile is placed without wrinkles. A geotextile should not be placed on a milled surface, but if it is, special care should be taken to ensure that the geotextile is bonded to the surface. Care should be taken in treating existing cracks if the geotextile is placed on a surface that does not have a leveling layer. Cracks wider than ¼ inch should be blown out and filled with a sealant. Any distressed areas with base or subgrade failure should be replaced. Inadequate surface preparation can result in poor performance of the completed HMA overlay.

After surface preparation is complete, an asphalt membrane is applied using a calibrated distributor truck. See Chapter 12, "Asphalt in Pavement Maintenance" for a discussion of the proper operation of the distributor. Selection of the proper type and application rate, along with a uniform distribution of the membrane, is essential to the performance of the interlayer and thus the overlay. The manufacturer's recommendations should be carefully followed regarding the type and amount of asphalt that should be applied with a specific geotextile.

The quantity of asphalt binder applied varies with the type and supplier of the geotextile. The amount of tack coat applied also depends on the condition of the underlying surface. An aged, weathered surface requires a higher application rate than a new HMA leveling course. It is essential to apply the membrane uniformly across the surface. A common range of application rates suggested by manufacturers is 0.20–0.30 gallon/yard^2 (1.4 litres/meter^2). The membrane should extend beyond the edges of the paving geotextile by 2–3 inches (50–75 millimeters). The temperature of the asphalt binder should be high enough to allow uniform application. Special care must be taken during application of the membrane in the overlap zone when two passes of a distributor are used to obtain coverage over the entire width of the geotextile being placed. The membrane should be placed over a sufficient length such that the distributor can place a uniform quantity. Care should be taken to avoid placing the membrane too far ahead of the paving process. Small, variable-width areas should be applied with the hand spray boom on the back of the distributor to ensure proper coverage. Before placing the membrane, the surface should be swept to remove debris. If weather or equipment breakdown cause paving operations to be suspended, the surface should not be opened to traffic because of safety concerns. Uncovered geotextile can have very low skid resistance.

The geotextile should be placed as near to the distributor truck as possible (See Figure 13.4). Machine placement by an experienced crew is the preferred method. The geotextile should not be placed during a strong wind of more than 20 miles per hour (32 kilometers per hour) because the wind will cause kiting of the geotextile, which inhibits laydown and causes wrinkles in the geotextile. The geotextile laydown machine shown in Figure 13.4 holds a roll of geotextile on a spool and maintains a tension of about 14 pound per square inch (9.6 Newton per square centimeter). As the geotextile is pulled off the roll, the broom presses it into the tack coat.

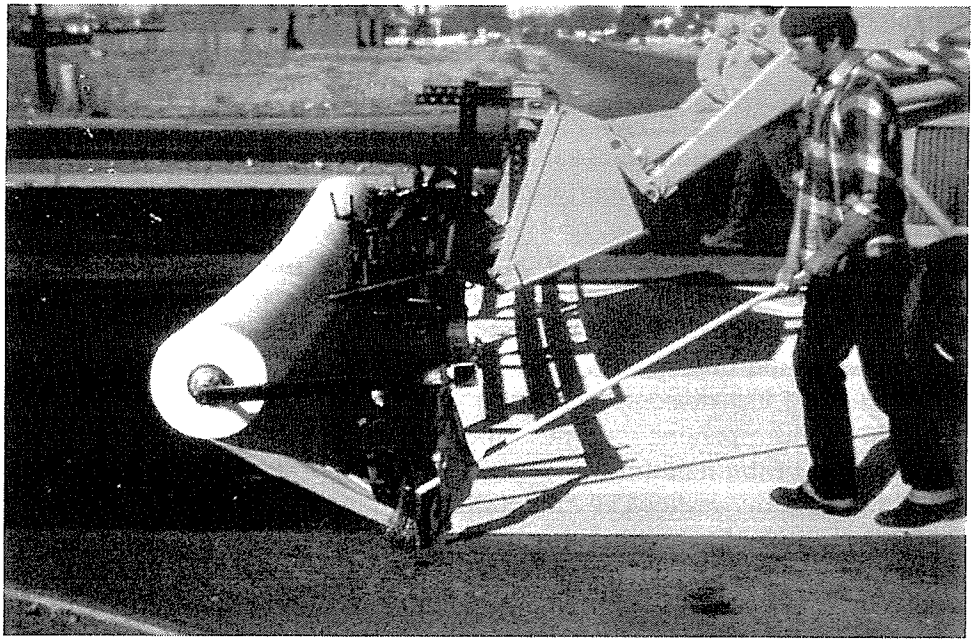


Figure 13.4 Placement of Paving Geotextile

It applies a uniform tension and pressure to the geotextile, resulting in a smooth surface without wrinkles. Wrinkles that develop during placement should be slit and folded over flat. If necessary, the wrinkles should be tacked down with either an asphalt binder or an asphalt emulsion. This process should also be used on curves. An overlap of 1–3 inches should be used between separate passes of the geotextile.

On overlays where a leveling course has been placed, it may be necessary to apply a light tack coat to the geotextile surface before placing the second layer. The HMA should be placed soon after the geotextile is placed. Two or three passes of a pneumatic roller is sometimes necessary to bond the geotextile to the old pavement (being careful not to roll up the fabric on the tires of the roller). Construction traffic may pull the asphalt through the geotextile, causing the geotextile to become saturated in the wheel paths. As a result, the truck tires may become covered with asphalt and pull up the geotextile. The contractor can remedy this by broadcasting a small amount of screenings or HMA in the wheel paths to reduce the adhesion to the tires and/or reduce the number of trucks operating on the geotextile. Also, if the paving machine begins to slip, the contactor can spread a small amount of HMA beneath the wheels or tracks on the paver to improve traction.

During inspection of the placement of the paving geotextile, the following items should be monitored:

- The asphalt membrane should be applied with a calibrated distributor, and the yield on the tack coat should be checked periodically.
- The geotextile should be placed in such a manner that it is smooth with no wrinkles.

- Blisters that develop under the geotextile before placement of the overlay should be rolled out with a pneumatic roller.
- The tack coat application and geotextile laydown should be coordinated to provide effective tacking of the geotextile to the existing surface.
- The joints should be properly overlapped.

Paving Membranes

Several manufacturers produce a composite membrane that consists of a paving geotextile saturated with asphalt and coated with a rubberized asphalt adhesive. The top surface is saturated with asphalt cement. Beneath the geotextile is a second layer—rubberized asphalt adhesive. The adhesive is protected by a release sheet that is removed just before placement. The paving membranes are basically a peel-and-stick type of system. It may be necessary to apply a primer to improve adhesion before placement of the membrane. This step is needed whenever problems with adhesion may develop, such as on a surface that is wet or dusty, where temperatures are cool, or on an irregular surface (for example, a milled surface). The prime coat should always be used before installing a paving membrane on a bridge deck or when the paving membrane will be exposed to moisture combined with traffic before placement of the overlay. Before a paving membrane is installed, the manufacturer should be consulted for proper installation procedures.

Strip Interlayers

Strip interlayers (Figure 13.5) are membrane materials designed to repair localized pavement cracking and pavement joints. As with surface-wide interlayers, these materials absorb and dissipate movement (strain) in the pavement, reduce reflection cracking, and minimize water infiltration. Strip interlayers usually consist of modified asphalt sandwiched between non-woven paving geotextile and high-modulus woven geotextile. The non-woven geotextile layer ensures a strong bond with the pavement surface. The center modified-asphalt layer provides stress relief and resistance to water infiltration. The top woven geotextile adds modulus and strength to resist cracking in the overlay. Strip interlayers come in 12- to 20-inch rolls that are 65 feet long.

The construction process is similar to that for paving geotextiles, as previously discussed:

- An asphalt cement tack coat of 0.1–0.2 gallon/yard^2 (0.45–0.9 litre/meter^2) is applied. For local or small applications, the tack coat can be hand sprayed. Over larger areas, a distributor truck can be used to ensure uniform distribution.
- The membrane is cut and placed immediately after applying the tack coat.
- The membrane is rolled in place with a hand-held pneumatic roller or with a vehicle tire to smooth the membrane and increase the bond.
- After the membrane has been placed, it can be immediately paved over. A tack coat is placed over the membrane and pavement at the

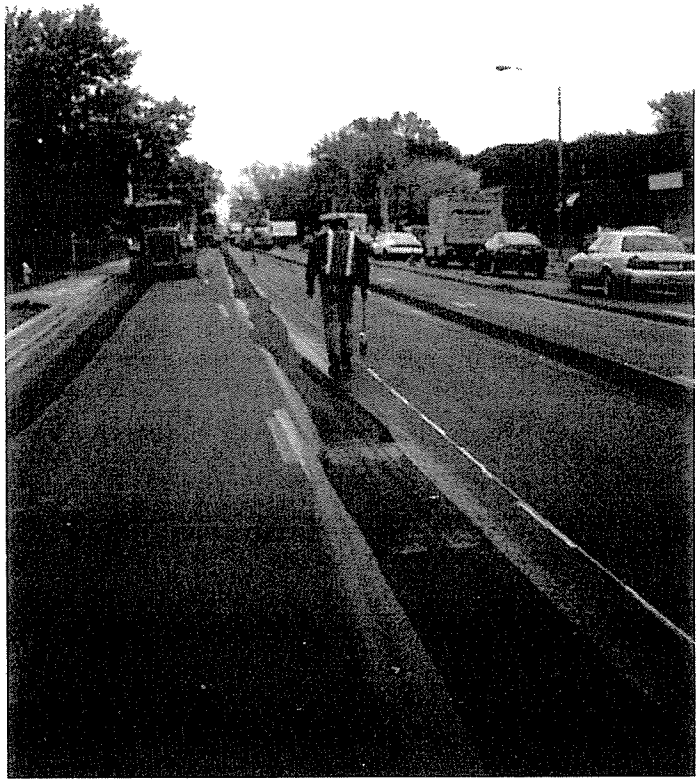


Figure 13.5 Placement of Strip Membrane

standard tack-coat rate. The overlay thickness should be based on anticipated traffic and should be a minimum of thickness of 1½ inches (37.5 millimeters).

Chip Seals

Chips seals are used as interlayers to help seal the underlying pavement and help bond the overlay to it, as well as a means to delay reflection cracking. These applications range from "conventional" single chip seals to multiple course treatments that have a similar effect as an open-graded crack relief layer in delaying reflection cracking. A chip seal may be placed on an existing pavement or on a milled surface. Placement procedures are described in Chapter 12, "Asphalt in Pavement Maintenance".

Stress-Absorbing Membrane

An asphalt-rubber stress-absorbing membrane interlayer (SAMI) is a chip seal that consists of a heavy, hot-applied [(325–425°F) (162–218°C)] crumb rubber (Figure 13.6) modified asphalt binder membrane, into which a relatively large cover aggregate is imbedded. The SAMI is very flexible and has a low modulus, allowing it to dampen movement at underlying cracks, reducing tensile stress in the HMA overlay. The membrane will also provide a waterproofing layer to reduce moisture intrusion into the underlying layers.

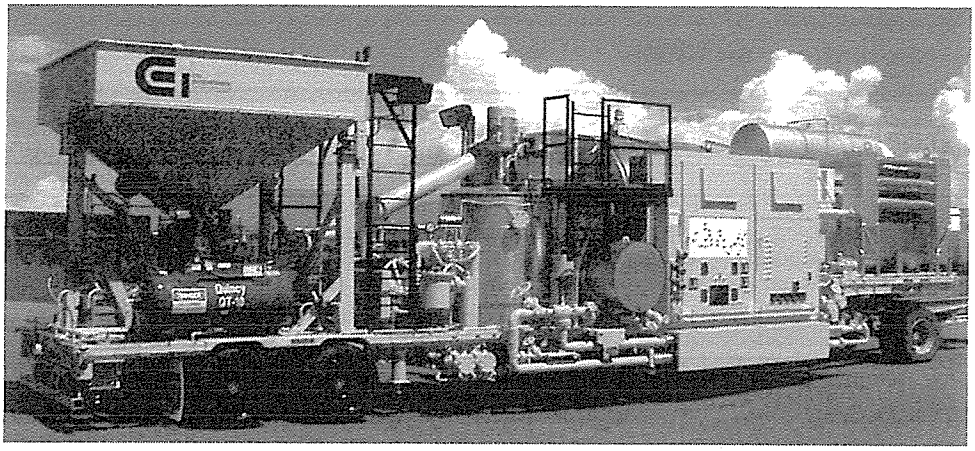


Figure 13.6 Onsite Asphalt Rubber Blending Equipment

Asphalt-rubber is defined by the (ASTM) D8 as: "a blend of asphalt cement, reclaimed tire rubber, and certain additives in which the rubber component is at least 15 percent by weight of the total blend and has reacted in the hot asphalt cement sufficiently to cause swelling of the rubber particles." Information about the production and qualities of asphalt-rubber can be found in Chapter 2, "Petroleum Asphalt," and from the Rubber Pavements Association (www.rubberpavements.org).

Asphalt-rubber is viscous and includes a dispersion of particulates that require application at higher rates and temperatures than other asphalt binders. Specially modified distributors that have larger and more powerful circulating pumps, larger heating systems, and larger spray nozzle openings are used.

Construction Process

The chip-seal construction procedures discussed in Chapter 12, "Asphalt in Pavement Maintenance" also apply to the construction of a SAMI. The hot rubber modified asphalt is sprayed with a properly calibrated distributor. It is spray-applied on an existing pavement at the rate of 0.6–0.8 gallon/yard^2 (2.7–3.6 litre/meter^2). The binders should be placed when the air temperature is above 70°F (21°C) and rising and the pavement surface is dry.

The liquid membrane is then covered with ⅜-inch (4.75 millimeter) or ¾-inch (12.5 millimeters) one-size aggregate at the rate of 30–35 pounds/yard^2. As with any hot-applied chip seal, it is imperative to use clean, dry cover stone to promote adhesion between the binder and aggregates. The cover stone can be precoated with 0.5–1.5 percent by mass residual asphalt to improve adhesion by removing the surface dust and keeping moisture from being absorbed into the chips. Alternately, some agencies require that the chips be delivered hot to the project. Table 13.3 provides the aggregate gradation for the construction of a SAMI.

Table 13.3
Gradations for SAMIs

Sieve Size	Gradation Used in (Percent passing)			
		California		
	Arizona	⅜ in. 9.5 mm	½ in. 12.5 mm	Florida
1 in. (25 mm)				100
¾ in. (19 mm)			100	90–100
½ in. (12.5 mm)		100	95–100	20–55
⅜ in. (9.5 mm)	100	70–85	70–85	0–15
¼ in. (6.25 mm)	0–15	–	–	–
No. 4 (4.75 mm)	100	0–15	0–15	0–5
No. 8 (2.36 mm)	0–5	0–5	0–5	–
No. 200 (.075 mm)	0–0.2	0–1	0–1	–

Inspection

During construction, the following should be checked:

- **The gradation of aggregate.** Samples should be taken at least once every 500 tons (453 tonnes) to ensure that the gradation requirements are being meet.
- **The viscosity of the asphalt-rubber binder.**
- **The aggregate spread rate.** The aggregate spreader should be calibrated. This can be done by placing a section of canvas of known size and passing the spreader over it while spreading the aggregate. The aggregate on the canvas is weighed and the spread rate calculated. Also during production, the yield should be checked by measuring a distance and noting how much aggregate is used to cover that length of roadway. Knowing the length and width of the roadway and the amount of aggregate used, it is possible to check the spread rate.
- **Embedment of aggregate.** Determine the embedment of the aggregate a day or so after construction of the chip seal. Remove several of the largest stones and determine the percentage of embedment of the stones into the asphalt rubber binder. The goal is 50–70 percent embedment. Additional pneumatic tire rollers may be necessary to ensure proper embedment before the liquid membrane cools.
- **The shot rate of the asphalt-rubber binder.** Most modern distributors have computer-controlled measuring systems.

13.3 Mill and Inlay

Cold milling a pavement consists of removing the surface to a specified depth with specially designed equipment. Cold-milling equipment is available in many sizes. The equipment will range from large high-capacity machines (see Figure 13.7) capable of milling up to a width of 16 feet. (4.9 meters) and a depth of 12 inches (300 millimeters) to smaller machines that can mill around manholes, valves, etc. (see Figure 13.8).

Cold milling is used to:

- Remove a deteriorated pavement to a desired depth, eliminating failed or faulty materials;
- Restore the pavement surface profile;
- Restore or maintain the drainage flow and curb reveal along street curbs;
- Provide a textured surface for skid resistance and improved bonding of an asphalt overlay; and/or
- Remove material under overhead structures to maintain clearances for bridge structures, traffic signals, and overhead utilities.

The byproduct of the milling operation, Reclaimed Asphalt Pavement (RAP), can be readily recycled into future asphalt pavements, saving both asphalt cement and aggregates. Processes for recycling RAP are described elsewhere.

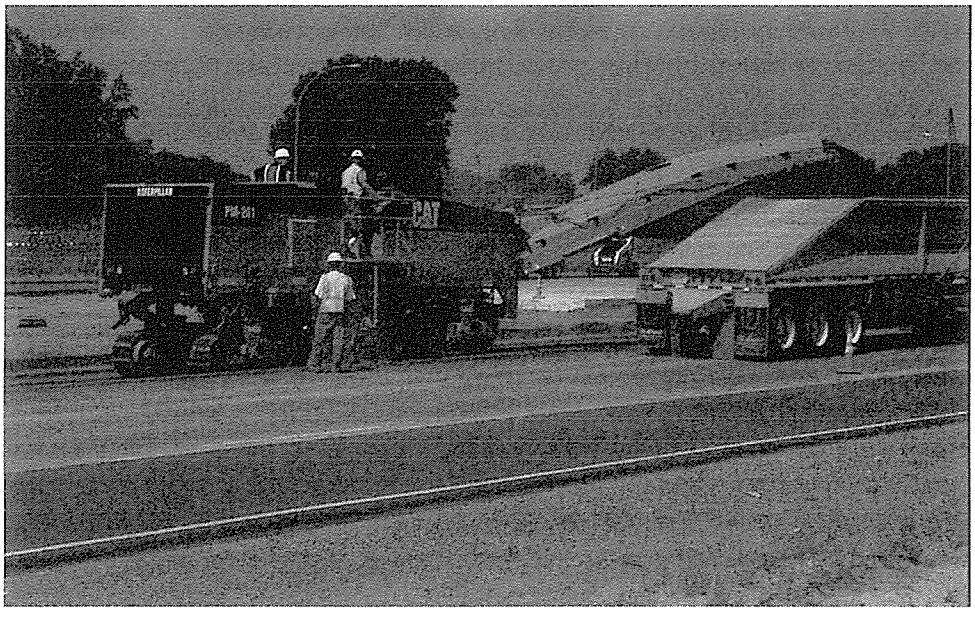


Figure 13.7 Large High Capacity Milling Machine

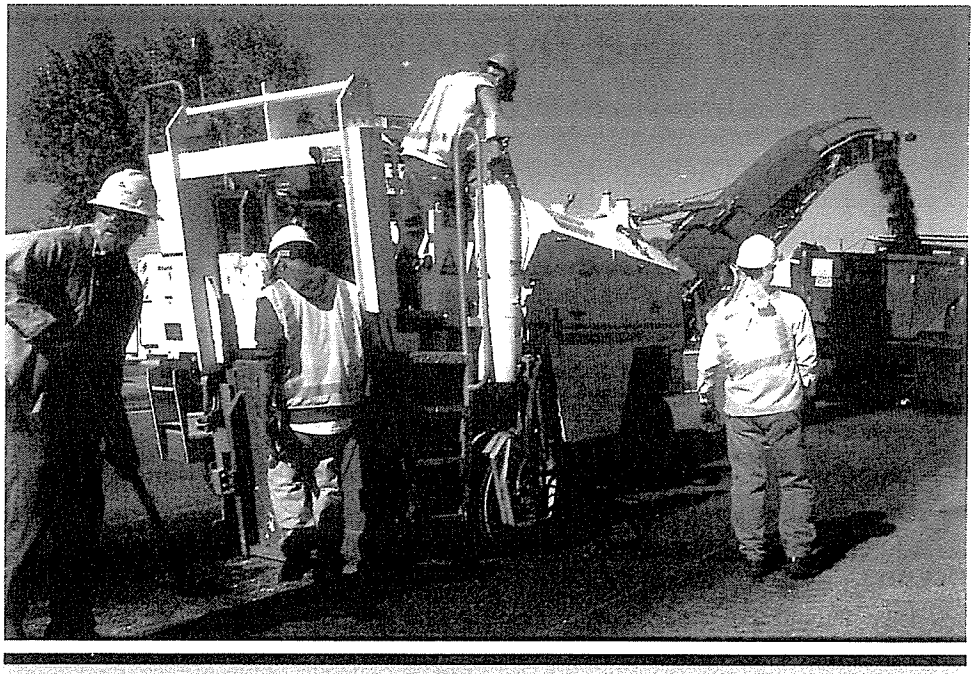


Figure 13.8 Medium Sized Milling Machine

Milling is conducted using a cutting drum, which is available in multiple widths. The cutting drum uses replaceable tungsten carbide teeth to cut the pavement surface (see Figure 13.9). These drums generally operate in the "upcutting" direction to facilitate breaking and removal of the existing material. The teeth are oriented in a helical or a spiral pattern to move the milled material to the center of the cutting drum, where it is forced onto a loadout conveyor for removal. The service life of the tungsten teeth depends on the nature and the depth of the material being milled. A small amount of water is used to control the amount of dust created by the milling process and to cool the cutting teeth to extend their life. The size of the reclaimed material can be adjusted by changing the rotational speed and/or reducing the forward speed of the milling machine.

Micromilling (also called fine milling) is a variation of the standard milling procedure. The micromilling process uses more cutting teeth to produce a much finer textured surface (see Figure 13.10). A conventional milling machine has teeth spaced ⅝ inch (15 millimeters) – ¾ inch (20 millimeters) apart. The teeth on a micromilling drum are typically spaced about 5/16 inch (8 millimeters) apart. This results in almost double the number of teeth for micromilling versus conventional milling. It is also possible to have the teeth spaced even closer at about ¼ inch for micromilling, but this is not very common. Micromilling can provide a very fine, uniform texture that can serve as the final riding surface with or without an overlay. It can also be used to correct some minor pavement distress or to remove pavement markings because of changing traffic patterns. With micromilling, the depth of cut is limited to about 1.5 inches (40 millimeters) because

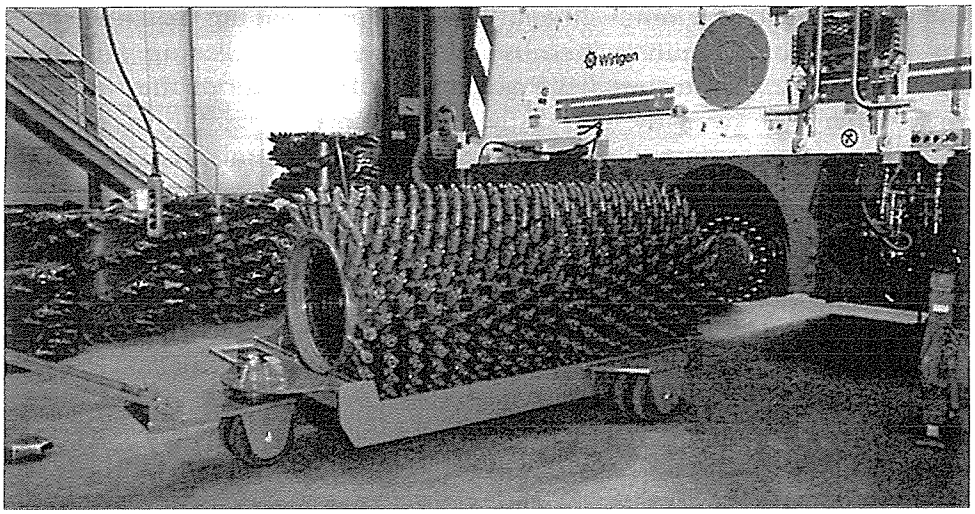


Figure 13.9 Tungsten Carbide Milling Teeth



Figure 13.10 Micromilled Surface

the extra teeth in contact with the pavement greatly increase the power needed to turn the cutting head.

A scraper blade is placed behind the cutting drum to collect the reclaimed material and to shave high points from the milled surface. The scraper blade and loading conveyor remove most of the milled material and load

it into haul trucks for removal from the site. Front-loading machines are now standard in the industry because they are easier to clean up after and safer to operate. The material is loaded over the tailgate of the removal truck, rather than over the cab of the truck, if a rear loading machine is used. This configuration allows the machine operator and truck driver to see each other, which helps communication. Also, because the operator can see into the bed of the removal truck, he can better judge the amount and distribution of material that he is loading into each truck. On some smaller milling machines, the reclaimed material is rear loaded or left on the pavement for the clean-up crew. The fine, loose particles that remain on the roadway after milling can be removed with power brooms in rural areas, but vacuum sweepers or power sweepers must be used in urban areas.

Using a milling machine can improve the ride of the final surface. Milling machines may use automatic leveling systems that operate with a number of different sensors similar to the ones found on paving machines.

13.3.1 Specifications

The end-result specification for cold milling is generally simple and includes provisions that address the following details:

- **A general requirement about the cold-milling machine.** A typical specification requires the contractor to use a machine with automatic grade and slope controls, capable of cold milling an existing pavement surface to the depth or profile specified.
- **The milling width and depth.** Whether to specify a depth of cut or a nominal depth depends on whether or not the milling operation is considered to be an opportunity to improve the profile, or ride quality, of the pavement. The smoothness, or ride quality, of the milled surface can only be obtained if a nominal depth of cut is specified
- **The maximum particle size of the reclaimed material.** This will generally be a maximum particle size of 1½–2 inches (37.5–50 millimeters).
- **The grade and slope requirements.** This will generally be included in the project plans or special provisions rather than in a standard specification.
- **The texture of the milled surface.** This can be a problem if the surface is open to traffic for an extended period before the overlay is placed.
- **Method of measure of quantities.** This can be either "volume (cubic yards [cubic meters]) or mass (tons [tonnes]) of reclaimed material removed," or "square yard (square meter) of surface milled" to a prescribed depth.
- **The ownership of the reclaimed material.** Some agencies retain ownership of some or all of the RAP generated from milling or other forms of reclamation, while others make it the property of

the contractor. When a specifier retains ownership, instructions for hauling and storing the millings must be included in the construction plans. Specifiers should consider that the reclaimed material is a valuable resource that can be used to reduce the cost of HMA construction while retaining the qualities of virgin HMA.

13.3.2 Inspection

During the cold-milling process, the specifying agency should monitor the following areas:

- The depth of the milling and the grade and cross-slope. This should include checking both the edge of the surface being milled and the crown. (Many times, the milling is used to correct the cross-slope of the roadway.)
- The moisture content of the reclaimed material, if the cold-milling is being paid for by mass.
- The smoothness of the surface, if a nominal depth of cut is specified. This is usually checked with a 10-foot. straightedge, a profilograph, or an inertial profiler as a part of a ride quality specification.